

force. Let m denote the total vehicle mass, g the gravitational constant, c_r the rolling friction coefficient (which depends on tire pressure, road surface conditions and other factors) and α the gradient angle. Then the rolling resistance F_r can be determined as

$$F_r = c_r \cdot m \cdot g \cdot \cos(\alpha).$$

With v denoting the speed, c_d the aerodynamic drag coefficient, ρ_a the air density and A_f the frontal area of the vehicle, the aerodynamic resistance is:

$$F_a = \frac{1}{2} \cdot \rho_a \cdot A_f \cdot c_d \cdot v^2.$$

Adding the gravitational force $F_g = m \cdot g \cdot \sin(\alpha)$, the total mechanical power P_M is:

$$P_M = \left(m \cdot a + \frac{1}{2} \cdot c_d \cdot \rho \cdot A \cdot v^2 + m \cdot g \cdot \sin(\alpha) + c_r \cdot m \cdot g \cdot \cos(\alpha) \right) \cdot v, \quad (1)$$

where a denotes the acceleration. As mentioned above, a is assumed to be zero in the following as we do not consider acceleration and braking processes.

In the second step, the electric power P_E that is needed to achieve the required mechanical power P_M is calculated, taking into account energy losses that occur in the electric engine. To this end, we use the Quantized State System (QSS) model of an electric engine presented in (Guzzella & Amstutz, 2005). The authors provide values for the engine efficiency at given pairs of torque and rotational velocity. We directly convert torque and rotational velocity to mechanical power. Using homogeneous regression, i.e. a linear regression with a y-axis intercept of zero, on the converted values, we obtain the following relationship between P_M and the discharged electric energy P_E^d (respectively the recuperated electric energy P_E^r) that is described by the regression coefficient ϕ :

$$\begin{aligned} P_E^d &= \phi^d \cdot P_M \text{ for } 0 \text{ kilowatts} \leq P_M \leq 100 \text{ kilowatts} \\ P_E^r &= \phi^r \cdot P_M \text{ for } -100 \text{ kilowatts} \leq P_M < 0 \text{ kilowatts.} \end{aligned}$$

The determined relationship is adequate for a mechanical power demand of up to 100 kilowatts, which we consider a realistic threshold for commonly employed ECV models.

In the third step, we compute the battery power P_B that is necessary to obtain the required electric energy P_E . For this purpose, we use the relationship for converting electric power P_E into battery power P_B for a lithium-ion accumulator presented in van Keulen, de Jager, Serrarens, and Steinbuch (2010). The presented relationship does not account for external effects (e.g., the ambient temperature) or the influence of the current charge level of the battery. We approximate the given relation by means of a homogeneous regression and obtain as final equations for discharging (P_B^d) and recuperating (P_B^r)

$$\begin{aligned} P_B^d &= \varphi^d \cdot P_E \text{ for } P_E \geq 0 \text{ kilowatts} \\ P_B^r &= \varphi^r \cdot P_E \text{ for } P_E < 0 \text{ kilowatts,} \end{aligned}$$

with the regression coefficient φ that describes the battery efficiency.

To compute energy consumption on a graph, vehicle mass is defined as a function of the currently loaded amount of cargo u , i.e., $m(u) = m_c + m_u \cdot u$, where m_c is the curb mass of the vehicle and m_u the mass of one unit of cargo. Let u_j denote the amount of cargo when arriving at customer j (before unloading), i.e., the amount required to fulfill the demand at customer j and all following customers. Further, let $P_{ij}(u_j)$ denote the constant mechanical power demand over the course of an arc (i, j) with a load that is equivalent to the remaining customer demand:

$$P_{ij}(u_j) = \left(\frac{1}{2} \cdot c_d \cdot \rho \cdot A \cdot v^2 + m(u_j) \cdot g \cdot (\sin(\alpha_{ij}) + c_r \cdot \cos(\alpha_{ij})) \right) \cdot v_{ij}$$

Then, the battery power demand on this arc is given by $P_B(P_E(P_{ij}(u_j)))$. Multiplying with the associated travel time t_{ij} of the arc yields the electric energy consumption b_{ij} for traveling this particular arc:

$$b_{ij}(u_j) = \begin{cases} \phi^d \cdot \varphi^d \cdot P_{ij}(u_j) \cdot t_{ij} & \text{if } P_{ij}(u_j) \geq 0 \text{ kilowatts} \\ \phi^r \cdot \varphi^r \cdot P_{ij}(u_j) \cdot t_{ij} & \text{if } P_{ij}(u_j) < 0 \text{ kilowatts.} \end{cases} \quad (2)$$

All variables except the amount of cargo u_j are constant for a given arc. Therefore, it is possible to split $b_{ij}(u_j)$ into a fixed and a load-dependent part and precompute parts of the expression.

3.2. Energy consumption of combustion engines

Mechanical power can be directly translated to diesel fuel consumption as shown in Demir et al. (2012). The fuel rate is given by $FR = \frac{\xi}{\kappa \cdot \psi} (kND + \frac{P_M}{\eta \cdot \eta_{tf}})$, where P_M is the mechanical power as defined in Eq. (1), ξ the fuel-to-air mass ratio, κ the heating value of typical diesel fuel, k the engine friction factor, N the engine speed, D the engine displacement, ψ a factor converting the fuel rate from grams per second to liters per second, η the efficiency parameter for diesel engines and η_{tf} the drive train efficiency. We assume that driving downhill in gear can reduce the engine consumption down to 0. The constant fuel rate over the course of an arc is given by:

$$FR_{ij}(u_j) = \max \left(\frac{\xi}{\kappa \cdot \psi} \left(kND + \frac{P_{ij}(u_j)}{\eta \cdot \eta_{tf}} \right), 0 \right).$$

Thus, the fuel consumption of an ICCV traversing an arc (i, j) with cargo u_j can be calculated as:

$$f_{ij}(u_j) = t_{ij} \cdot FR_{ij}(u_j). \quad (3)$$

4. The Electric Vehicle Routing Problem with Time Windows and Mixed Fleet (E-VRPTWMF)

We define E-VRPTWMF on a complete, directed graph $G = (V'_{0,N+1}, A)$. $V = \{1, \dots, N\}$ denotes the set of customers, F the set of recharging stations, F' the set of visits to vertices in F and V' denotes the union of V and F' . Vertices 0 and $N+1$ denote instances of the depot, and all routes start at 0 and end at $N+1$. Inclusion of one or both depot instances is indicated by subscripting the respective set, e.g., the set $V'_{0,N+1}$ contains both instances of the depot. The set of arcs is given by $A = \{(i, j) \mid i, j \in V'_{0,N+1}, i \neq j\}$.

With each vertex $i \in V'_{0,N+1}$, we associate a nonnegative demand q_i ($q_i = 0$, $i \notin V$), a nonnegative service time s_i ($s_i = 0$, $i \notin V$) and a time window $[e_i, l_i]$, within which service at the customer has to start. Time windows are hard, i.e., starting service late is not allowed but waiting in case of early arrival is possible. The time window $[e_0, l_{N+1}]$ at the depot specifies the scheduling horizon of the problem. Each arc is described by the following properties: distance d_{ij} , travel speed v_{ij} , travel time $t_{ij} = d_{ij}/v_{ij}$ and a factor α_{ij} representing the gradient of the terrain.

A mixed vehicle fleet of fixed size consisting of m_E ECVs and m_{IC} ICCVs is positioned at the depot. All vehicles have the same maximal cargo loading capacity Q . On each visit to a recharging station, ECVs are recharged to their maximum battery capacity B . The recharging time depends on the recharging rate r and the difference between B and charge level on arrival at the station. Electricity and fuel consumption of a vehicle traveling an arc are determined by the models introduced in Section 3.

Thus, E-VRPTWMF can be formulated as a (nonlinear) mixed-integer program. For every arc $(i, j) \in A$, we define the following binary decision variables: Variables x_{ij}^E are equal to 1 if an arc (i, j) is traveled by an ECV, variables x_{ij}^{IC} if the arc is traveled by an ICCV. Otherwise, the decision variables are 0. Variables τ_i define the arrival time, u_j the remaining cargo (before unloading) and y_i the remaining charge level on arrival at vertex $i \in V'_{0,N+1}$. Table 1 summarizes the variables and parameters of our model.

Table 1
Variable and parameter definitions of the E-VRPTWMF model.

$0, N+1$	Depot instances
F'	Set of visits to recharging stations, dummy vertices of the set of recharging stations F
V	Set of customer vertices $V = \{1, \dots, N\}$
V'	Set of customers and recharging visits: $V' = V \cup F'$
d_{ij}	Distance between vertices i and j
t_{ij}	Travel time between vertices i and j
Q	Vehicle capacity
r	Recharging rate
B	Battery capacity
$b_{ij}(u_j)$	Electric energy consumption between between vertices i and j depending on cargo u_j (Eq. (2))
$f_{ij}(u_j)$	Fuel consumption between between vertices i and j depending on cargo u_j (Eq. (3))
q_i	Demand of vertex i , 0 if $i \notin V$
e_i	Earliest start of service at vertex i
l_i	Latest start of service at vertex i
s_i	Service time at vertex i ($s_0, s_{N+1} = 0$)
τ_i	Decision variable specifying the time of arrival at vertex i
u_i	Decision variable specifying the remaining cargo on arrival at vertex i
y_i	Decision variable specifying the remaining battery capacity on arrival at vertex i
x_{ij}^E	Binary decision variable indicating if arc (i, j) is traveled by an ECV
x_{ij}^C	Binary decision variable indicating if arc (i, j) is traveled by an ICCV

In this work, we consider three alternative objective functions for E-VRPTWMF:

1. Minimize traveled distance f_d :

$$f_d = \sum_{i \in V_0, j \in V'_{N+1}, i \neq j} d_{ij} (x_{ij}^E + x_{ij}^C) \rightarrow \min \quad (4)$$

2. Minimize cost f_c , consisting of the cost for vehicle propulsion and labor cost. Cost factors c^E and c^C denote the cost of electric energy, respective diesel fuel and c^D the driver wage per time unit:

$$f_c = \sum_{i \in V_0, j \in V'_{N+1}, i \neq j} [c^E \cdot b_{ij}(u_j) \cdot x_{ij}^E + c^C \cdot f_{ij}(u_j) \cdot x_{ij}^C] + \sum_{i \in V'} c^D (x_{iN+1}^E + x_{iN+1}^C) (\tau_i + s_i + t_{iN+1}) \rightarrow \min \quad (5)$$

3. Minimize cost including cost for battery replacement f_{cb} . The distance traveled by ECVs is multiplied by a cost factor c^B that accounts for the cost of battery replacement. It is expressed as cost per kilometer by dividing acquisition cost by the expected total lifetime mileage.

$$f_{cb} = f_c + \sum_{i \in V_0, j \in V'_{N+1}, i \neq j} c^B \cdot d_{ij} \cdot x_{ij}^E \rightarrow \min \quad (6)$$

The constraints of E-VRPTWMF are as follows:

$$\sum_{j \in V'} (x_{ij}^E + x_{ij}^C) = 1 \quad \forall i \in V \quad (7)$$

$$\sum_{j \in V_{N+1}, i \neq j} x_{ij}^E \leq 1 \quad \forall i \in F' \quad (8)$$

$$\sum_{j \in V_{N+1}, i \neq j} x_{ij}^E - \sum_{j \in V_0, i \neq j} x_{ji}^E = 0 \quad \forall i \in V' \quad (9)$$

$$\sum_{j \in V_{N+1}, i \neq j} x_{ij}^C - \sum_{j \in V_0, i \neq j} x_{ji}^C = 0 \quad \forall i \in V \quad (10)$$

$$\sum_{j \in V'} x_{0j}^E \leq m_E \quad (11)$$

$$\sum_{j \in V'} x_{0j}^C \leq m_C \quad (12)$$

$$\tau_i + (s_i + t_{ij}) (x_{ij}^E + x_{ij}^C) - l_0 (1 - x_{ij}^E - x_{ij}^C) \leq \tau_j \quad \forall i \in V_0, \forall j \in V'_{N+1}, i \neq j \quad (13)$$

$$\tau_i + t_{ij} \cdot x_{ij}^E + r (B - y_i) x_{ij}^E - (l_0 + rB) (1 - x_{ij}^E) \leq \tau_j \quad \forall i \in F', \forall j \in V'_{N+1}, i \neq j \quad (14)$$

$$e_i \leq \tau_i \leq l_i \quad \forall i \in V'_{0,N+1} \quad (15)$$

$$u_i - q_i (x_{ij}^E + x_{ij}^C) + Q (1 - x_{ij}^E - x_{ij}^C) \geq u_j \geq 0 \quad \forall i \in V_0, \forall j \in V'_{N+1}, i \neq j \quad (16)$$

$$0 \leq u_0 \leq Q \quad (17)$$

$$y_i - b_{ij}(u_j) \cdot x_{ij}^E + B (1 - x_{ij}^E) \geq y_j \quad \forall i \in V, \forall j \in V'_{N+1}, i \neq j \quad (18)$$

$$B - b_{ij}(u_j) \cdot x_{ij}^E \geq y_j \quad \forall i \in F_0', \forall j \in V'_{N+1}, i \neq j \quad (19)$$

$$B \geq y_i \geq 0 \quad \forall i \in V'_{0,N+1} \quad (20)$$

$$x_{ij}^E, x_{ij}^C \in \{0, 1\} \quad \forall i, j \in V'_{0,N+1}, i \neq j \quad (21)$$

Constraints (7) enforce that each customer visit has exactly one successor. Constraints (8) guarantee that each visit to a recharging station is used at most once, i.e., not all recharging visit vertices must be used. Flow conservation constraints (9) and (10) guarantee for each vertex that the number of incoming arcs is equal to the number of outgoing arcs for each vehicle type. Constraints (11) and (12) ensure that the maximum number of employed vehicles adheres to the fleet composition. Constraints (13) and (14) link arrival times at vertices i and j if the arc from i to j is traveled. Constraints (14) covers the case with recharging visits: Here, recharging times that depend on the remaining charge level y_i when arriving at station i have to be taken into account. Constraints (15) guarantee arrival within the time window at each vertex. Constraints (16) enforce the fulfillment of demand at customer vertices. Constraint (17) restricts the initial cargo load level to the maximum capacity of a vehicle.

Constraints (18) set the battery level at a vertex succeeding a customer visit in accordance with the energy consumption for discharging or recuperating. For vertices succeeding a visit to a recharging station or the depot, Constraints (19) set the battery level equal to the maximum battery capacity reduced by the energy required on the respective arc. Here, the battery is either discharged or recuperation takes place. Constraints (20) restricts the charge level to the maximum battery capacity in order to prevent recuperation beyond the maximal capacity. Finally, binary decision variables are defined in constraints (21).

Although our model covers several real-world aspects of ECVs, we still use some simplifications. Gradient and speed are considered fixed over the course of an arc. However, a more fine-grained topology and acceleration patterns can be integrated, e.g., by considering a path $p(i, j)$ instead of the arc (i, j) . A path $p(i, j)$ is defined by a fixed sequence